E-WASTE MANAGEMENT PRACTICES OF LEADING ELECTRONICS MANUFACTURERS IN INDIA: A COMPARATIVE STUDY OF RULES FOLLOWED AND PRACTICES NOT DIRECTLY IMPLEMENTED

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Abstract

Electronic waste (e-waste) has emerged as a major environmental and public-health concern because electrical and electronic equipment contains valuable materials as well as hazardous substances. The reference study by Chadha and Chadha examines the growth of e-waste, its health and environmental impacts, international initiatives and the legal framework for e-waste management in India. Building on that framework, the present study compares the publicly disclosed e-waste management practices of five electronics manufacturers selected from the provided list of leading electronics manufacturers in India: Dixon Technologies, Kaynes Technology, Syrma SGS Technology, Centum Electronics and Foxconn. The study identifies practices that are disclosed as being followed such as waste segregation, use of authorized recyclers, recycling/reuse, hazardous-waste controls, RoHS-related controls, and EPR-related mechanisms where applicable and separately identifies practices that are not directly implemented, not applicable to the company's business model, or not publicly quantified. The comparison shows that B2B electronics manufacturing models often rely on authorized downstream recyclers rather than direct end-of-life product take-back. The paper emphasizes that absence of a direct take-back programme should not automatically be interpreted as a legal violation. The study also highlights disclosure gaps, particularly where companies do not report product-level recovery data. The findings suggest that stronger traceability, transparent recovery metrics, life-cycle assessment and clearer communication of responsibilities across the electronics value chain can further improve circular e-waste management in India.

Keywords: E-waste, EPR, Recycling, Authorized Recyclers, Electronics Manufacturing, Circular Economy, RoHS, Waste Management

1. INTRODUCTION

Rapid technological development has increased the production, use and replacement of electrical and electronic equipment. The reference paper defines e-waste as electrical and electronic equipment and parts that have been discarded by their owner without an intention of reuse. It notes that technological products may contain large numbers of substances, some of which are toxic, making inappropriate disposal a significant environmental and health concern.

The reference study also describes e-waste as a rapidly growing waste stream and links its growth to increasing consumption of electrical and electronic equipment, short product life cycles and limited repair options. It reports that India was among the world's major e-waste-generating countries and emphasizes the need for formal collection, environmentally sound recycling and effective legal enforcement.

India's regulatory framework has evolved since the rules discussed in the reference paper. The E-Waste (Management) Rules, 2022 superseded the 2016 rules and came into force on 1 April 2023. The current framework uses an Extended Producer Responsibility (EPR) system and a CPCB online portal for relevant manufacturers, producers, refurbishers and recyclers. The rules focus on environmentally sound management, recycling targets and formalization of the e-waste value chain.

2. MATERIAL AND METHOD

This study follows the analytical approach of the reference paper while updating the company-level evidence. The reference paper is used as the conceptual basis for e-waste impacts, legal development, recycling challenges and recommendations. Company-level information is taken from publicly available annual reports, business responsibility and sustainability reports, official EPR pages and official sustainability disclosures. The five companies were selected from the electronics manufacturers shown in the supplied reference image.

The analysis uses two categories. First, 'Followed / Disclosed' refers to a practice that the company explicitly reports as implemented. Second, 'Not directly implemented / Not applicable / Disclosure gap' refers to a practice that the company explicitly says it does not perform, reports as not applicable, or for which the available disclosure does not provide sufficient evidence. These categories are deliberately not labelled 'legal violations' unless a source establishes such non-compliance.

The five companies examined are: Dixon Technologies (India) Limited, Kaynes Technology India Limited, Syrma SGS Technology Limited, Centum Electronics Limited and Foxconn (Hon Hai Technology Group). The study is comparative and based on public disclosures; therefore, it does not constitute an independent environmental audit.

3. LEGAL AND POLICY FRAMEWORK

The reference paper discusses the Basel Convention, WEEE and RoHS initiatives, international cooperation and India's earlier e-waste rules. It also explains that authorized collection and recycling are essential and that EPR can make manufacturers financially, physically and legally responsible for their products.

The current Indian framework is the E-Waste (Management) Rules, 2022 and subsequent amendments. The rules apply to notified electrical and electronic equipment and relevant entities involved in manufacture, production, refurbishing, dismantling and recycling. The CPCB EPR portal provides registration and recycling-target mechanisms for applicable producers and requires registered stakeholders to use the portal for EPR compliance. The framework also provides for environmental compensation in cases of non-compliance. [1][2]

Key regulatory principles relevant to this comparison include: (i) registration where applicable; (ii) EPR obligations for producers covered by the rules; (iii) channelization of e-waste through registered/authorized recycling mechanisms; (iv) environmentally sound storage, handling and recycling; (v) periodic returns and documentation where applicable; and (vi) audit and enforcement mechanisms. These requirements must be distinguished from voluntary sustainability practices such as zero-waste-to-landfill targets or life-cycle assessment.

4. COMPANY-WISE COMPARATIVE ANALYSIS

The following comparison identifies the practices that each company publicly reports as followed and the practices that are not directly implemented, not applicable or not sufficiently disclosed. The distinction is important because B2B electronics manufacturers may not control the end-of-life stage of branded products sold by their customers.

4.1 DIXON TECHNOLOGIES

Dixon's FY 2024-25 disclosure states that it partners with authorized third-party recyclers and does not directly receive end-of-

life products. It also reports Zero Waste to Landfill across manufacturing facilities and an e-waste recycling target for 2032. [3]

4.2 KAYNES TECHNOLOGY

Kaynes' latest annual-report disclosure describes EPR applicability, authorized recycling channels and the absence of a direct product take-back/reclamation programme. [4]

4.3 SYRMA SGS TECHNOLOGY

Syrma SGS explicitly states that it operates as a B2B EMS provider, does not reclaim products or packaging at end of life and relies on its authorized recycler for EPR collection, transportation, storage, segregation, dismantling and recycling. [5][6]

4.4 CENTUM ELECTRONICS

Centum's BRSR describes an established waste-management system and approved e-waste recycler, while its EPR applicability is reported as Not Applicable. [7][8]

4.5 FOXCONN

Foxconn reports qualified e-waste recyclers and India Zero Waste to Landfill validation, while its sustainability disclosure notes a data-compilation limitation for quantitative e-waste recovery. Its 2023 third-party audit reported 144 overall non-compliances but no significant environmental non-compliance. [9][10][11]

5. COMPARATIVE TABLE OF RULES / PRACTICES

Table 1 summarizes the evidence in a form suitable for comparison. 'Yes' means explicitly disclosed as followed; 'No direct' means the company does not directly perform the activity; 'N/A' means the company reports the requirement as not applicable; and 'Gap' means the source reviewed does not provide complete evidence or quantitative disclosure.

6. FINDINGS AND DISCUSSION

All five manufacturers disclose formal waste-management measures, but the level and type of responsibility differ. A common pattern is the use of authorized or approved recyclers instead of a company-operated end-of-life collection and take-back network. This is particularly visible in B2B/EMS business models, where manufacturers produce for brand owners and may not sell directly to final consumers.

Second, source segregation, reuse/recycling and regulated downstream disposal are common practices. Dixon and Kaynes explicitly describe 3R-based systems; Syrma SGS emphasizes reduce/re-use, ISO 14001 and RoHS/REACH-compliant materials; Centum reports segregation and authorized disposal; and Foxconn reports qualified recycling vendors and zero-waste initiatives.

Third, EPR applicability is not identical across the five companies. Kaynes reports EPR as applicable, while Centum reports EPR as not applicable. Syrma SGS's disclosures emphasize its B2B EMS role and use of an authorized recycler for EPR activities. Therefore, EPR status should be interpreted from each company's legal role and product responsibility rather than assuming that every electronics manufacturer has identical producer obligations.

Fourth, the main gaps identified in this comparative study are direct product take-back, end-of-life reclamation, complete life-cycle assessment and transparent quantitative reporting. Dixon and Kaynes explicitly report no direct take-back/reclamation programme; Syrma SGS reports no product reclamation and limited control over finished-goods end-of-life; Dixon reports no LCA; and Foxconn's cited disclosure notes that quantitative e-waste recovery data was not yet fully available.

Finally, the reference paper stresses that India needs stronger enforcement, improved infrastructure, awareness, better collection and recycling practices, and integration of the organized and informal sectors under safe conditions. The company comparison supports this broader conclusion: responsible downstream recycling is necessary, but traceability across the full product life cycle is equally important.

7. RECOMMENDATIONS

1. Establish measurable end-of-life recovery indicators, even for B2B manufacturers, so that the downstream fate of e-waste can be tracked.

2. Expand digital traceability of e-waste from generation/collection through authorized recycling and material recovery.

3. Where legally and commercially appropriate, develop take-back or reverse-logistics mechanisms with brand owners and authorized recyclers.

4. Conduct and publish Life Cycle Assessments for major product categories to identify environmental impacts before products reach the waste stage.

5. Publish e-waste-specific recovery and recycling quantities rather than only aggregate waste figures.

6. Maintain documented verification of recycler authorization and audit downstream waste processors periodically.

7. Improve consumer and customer awareness so that e-waste is not mixed with municipal waste or transferred to unsafe informal channels.

8. Continue reducing hazardous substances through RoHS/REACH-aligned design and safer material substitution.

9. Align company sustainability targets with the current E-Waste (Management) Rules, 2022 and applicable EPR requirements, rather than relying only on voluntary zero-waste targets.

10. Improve coordination among manufacturers, brand owners, recyclers, regulators and consumers to create a traceable circular-economy model.

8. CONCLUSION

E-waste management in electronics manufacturing requires more than the disposal of factory-generated waste. It requires responsible handling throughout the product life cycle, including segregation, safe storage, authorized recycling, material recovery, hazardous-substance control and, where applicable, EPR obligations. The five-company comparison demonstrates that major electronics manufacturers have adopted several responsible practices, especially the use of authorized recyclers and recycling/reuse systems.

At the same time, important limitations remain. Direct product take-back is not implemented by several B2B manufacturers, some companies do not directly control end-of-life handling of finished goods, and quantitative e-waste recovery data or life-cycle assessments are not always publicly available. These limitations should be described as business-model constraints, non-applicability or disclosure gaps—not automatically as legal violations.

The reference paper concludes that effective legislation, monitoring, infrastructure, awareness and sustainable recycling technology are required to make the circular economy a reality.The present comparison supports that conclusion and adds a company-level perspective: the next stage of e-waste management in India should focus on measurable traceability, transparent recovery data, stronger reverse logistics and coordinated responsibility across the electronics value chain.

Table 1. Comparative status of major e-waste practices

Company	Segregation / 3R	Authorized recycler	EPR / regulatory mechanism	Direct take-back	Key gap
Dixon	Yes	Yes	Applicable regulatory compliance	No direct	No direct end-of-life reclamation; no LCA disclosed
Kaynes	Yes	Yes	EPR reported applicable	No direct	No direct take-back/reclamation
Syrma SGS	Yes	Yes	EPR through recycler	No direct	No control over finished-goods end-of-life; no all-state collection points
Centum	Yes	Yes	EPR reported	No direct	EPR N/A; no

			N/A	identified	external assurance in cited FY25 disclosure
Foxconn	Yes	Yes	Global/market-specific systems	Not established in reviewed India sources	Quantitative e-waste recovery data not fully disclosed; overall audit findings

Note: "Not directly implemented / N/A / gap" is not equivalent to legal non-compliance. Legal applicability depends on the entity's role under the E-Waste (Management) Rules, 2022 and the specific products and activities involved.

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